Optimising Evaluation of Arthematic Statements

Problem Statement

You are supposed to write 2 programs, one to parse the arthematic statements and output reduced arthematic statements

Another Program to parse reduced arthemtic statements and output a set of optimised reduced arthematic statements (reduced form) with meaning identical to the original arthematic expression that is, it has the identical output when interpreted.

You will be judged on the size of optimised reduced form and execution time of your program

1. Remove unnecessery definitions
2. Constant propagation

Input: Arthematic Statements

Output I : Reduced Arthematic Statements in form of Binary Operations and Print Statements

Output II : Optimised Output II

Input Format:

Multiple Arthematic Statements, each in a new line.

Structure of Arthamtic Statements:

VARIABLE = ARTH_EXPR print VALUE

ARTH_EXPR VALUE (BinOp VALUE) *BinOp is +, -, , /*

VALUE: INTEGER or VARIABLE

VARIABLE string of digits and charecters, not beginning with a digit

INTEGER sequence of digits

Reduced Arthematic Statements

VARIABLE = Value BinOp Value VARIABLE = Value print VALUE

Examples

Example 1

Input:

```
x = 2 * 3 + 5 - 1
```

Output I:

```
t1 = 2 * 3
t2 = t1 + 5
t3 = t2 - 1
x = t3
```

Output II:

Example 2 Input:

```
x = 2 + 3 * y - 1
print x
```

Output I:

```
t1 = 2*3
t2 = t1 + y
t3 = t2 - 1
x = t3
print x
```

Output II:

```
p1 = 6 + y
x = p1 - 1
print x
```